Markscheme

Mathematics: analysis and approaches

Practice question bank

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28. (a)

Consider the function $f(x) = \dfrac{x^2 - 2x - 57}{(x+3)^2(x-4)}$.

Write $f(x)$ as the sum of partial fractions in the form $\dfrac{A}{x+3} + \dfrac{B}{(x+3)^2} + \dfrac{C}{x-4}$.

attempt to write $A(x+3)(x-4) + B(x-4) + C(x+3)^2$ *(M1)*

$x^2 - 2x - 57 = A(x+3)(x-4) + B(x-4) + C(x+3)^2$ **A1**

substituting $x = -3$: $-42 = -7B \Rightarrow B = 6$ **A1**

substituting $x = 4$: $-49 = 49C \Rightarrow C = -1$ **A1**

comparing x^2 coefficients: $1 = A + C \Rightarrow A = 2$ **A1**

[5 marks]

(b)

Hence find the exact value of $\int_5^6 f(x)\,dx$.

$\int f(x)\,dx = 2\ln|x+3| - \dfrac{6}{x+3} - \ln|x-4|$ *(M1)*

evaluating between $x = 5$ and $x = 6$ gives ≈ -0.374 **A1**

[2 marks]